

Pretty Good Privacy (PGP)

What is PGP and how to use it

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2026-08-10

OtaNix ry 

PGP is the closest you're likely to get to military-grade encryption.

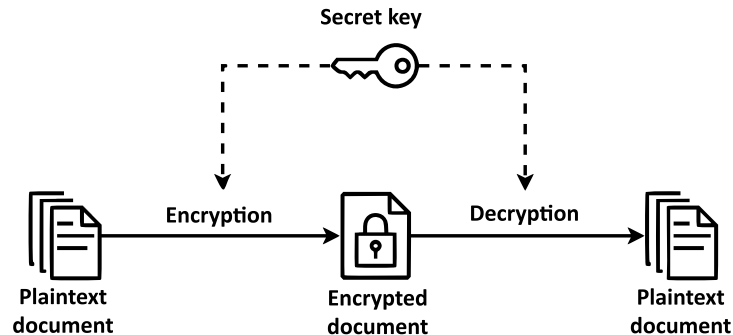
— Bruce Schneier

1. Symmetric vs. Asymmetric crypto
2. Keys, keypairs, and subkeys
3. Algorithms
4. PGP vs. GPG
5. Example: Using GPG
6. Hardware keys

Symmetric vs. Asymmetric crypto

Symmetric

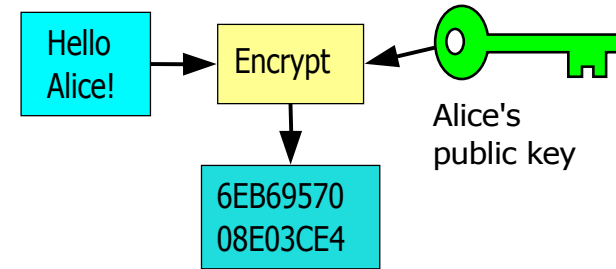
- (en|de)crypting uses the same key
- The entire key is private



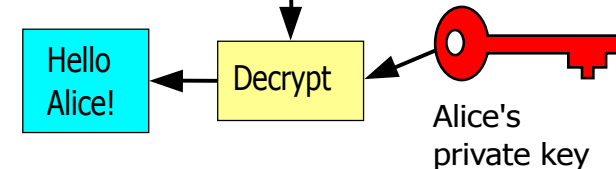
Asymmetric

- (en|de)crypting use different keys
- The key has a public and a private part

Bob



Alice



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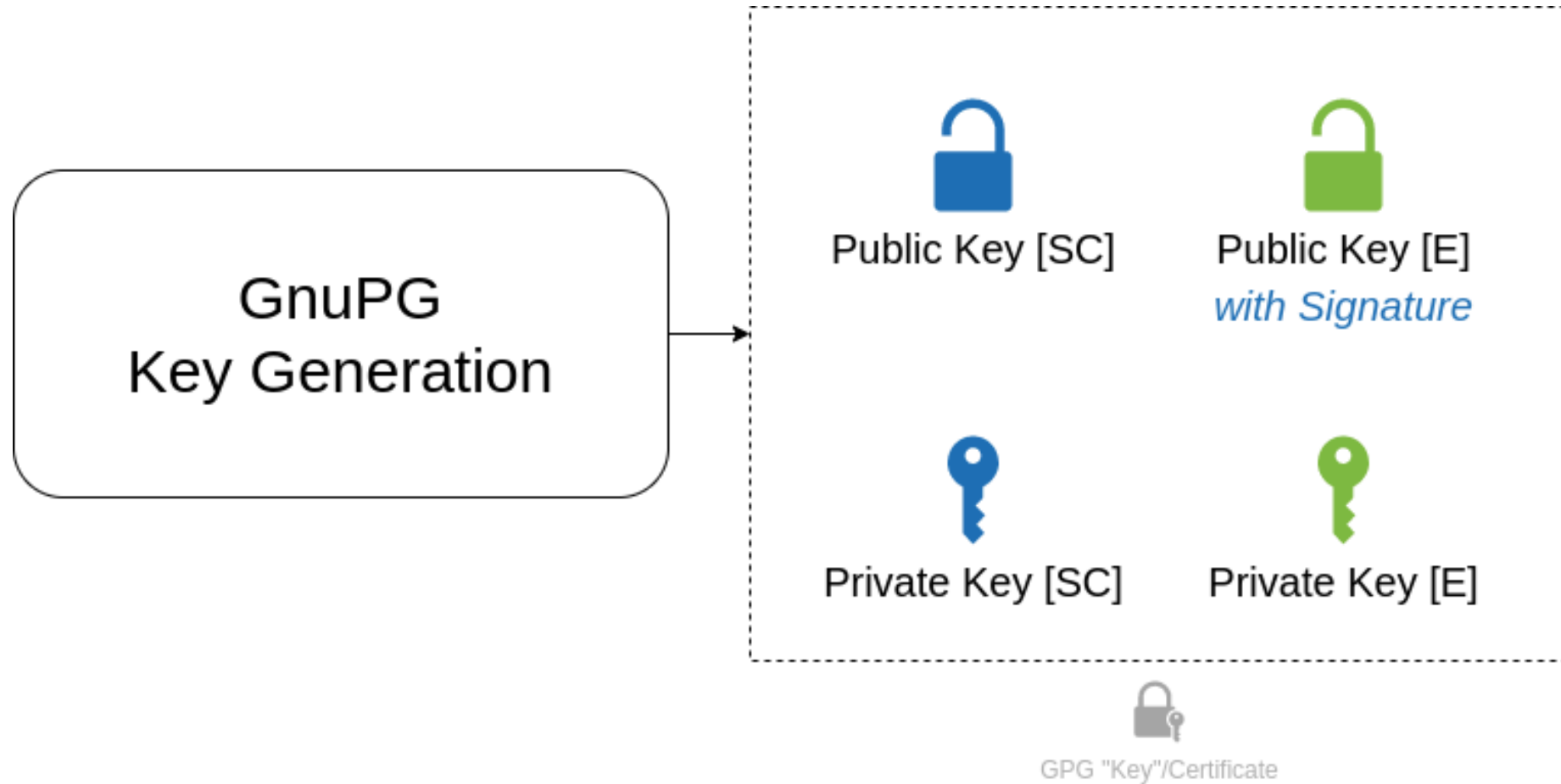
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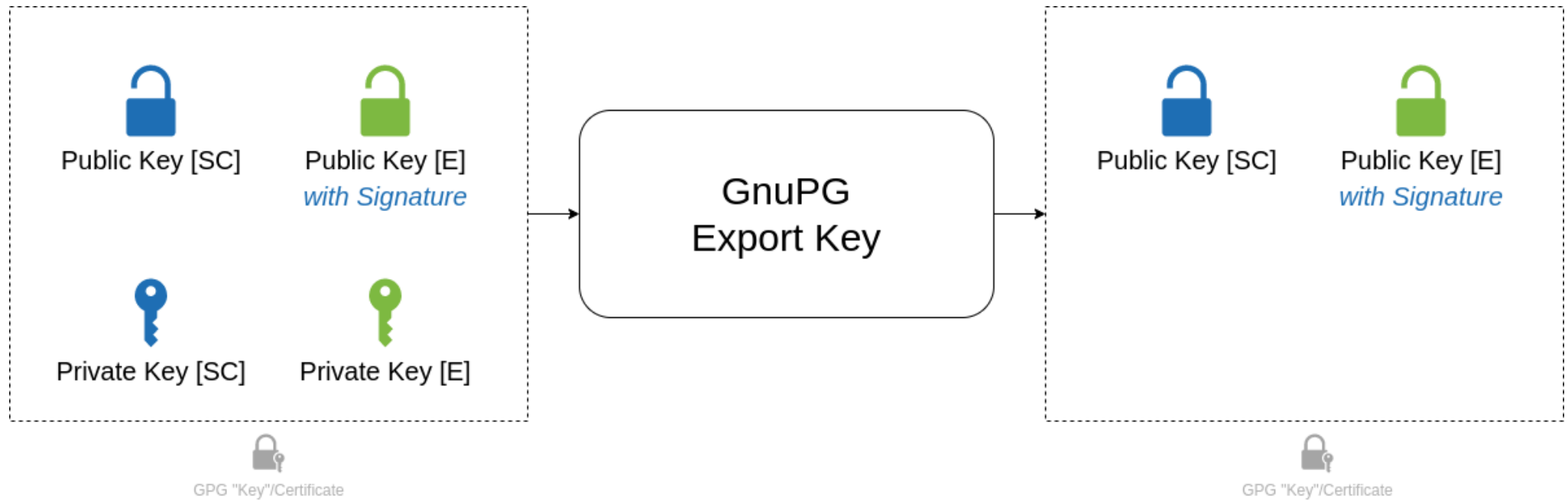
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- In PGP, a **key** (certificate) is really a collection of keypairs: a primary **certification** key plus subkeys, and identification metadata
- The word “key” can refer to any of the above (or even just a public key), but we use it only to refer to a PGP key

Keys, keypairs, and subkeys in PGP



Images from: <https://web.archive.org/web/20251027132314/https://rgoulter.com/blog/posts/programming/2022-06-10-a-visual-explanation-of-gpg-subkeys.html>

Keys, keypairs, and subkeys in PGP



A table of popular cryptographic algorithms

Algo	Type	Supported by YubiKey
AES	Symmetric	Yes*
ChaCha20	Symmetric	No
RSA	Asymmetric	Yes
ECDSA	Asymmetric	Yes
Ed25519	Asymmetric	Yes
X25519	Asymmetric	Yes

* AES is supported for PIV management-key operations.

How PGP encryption works

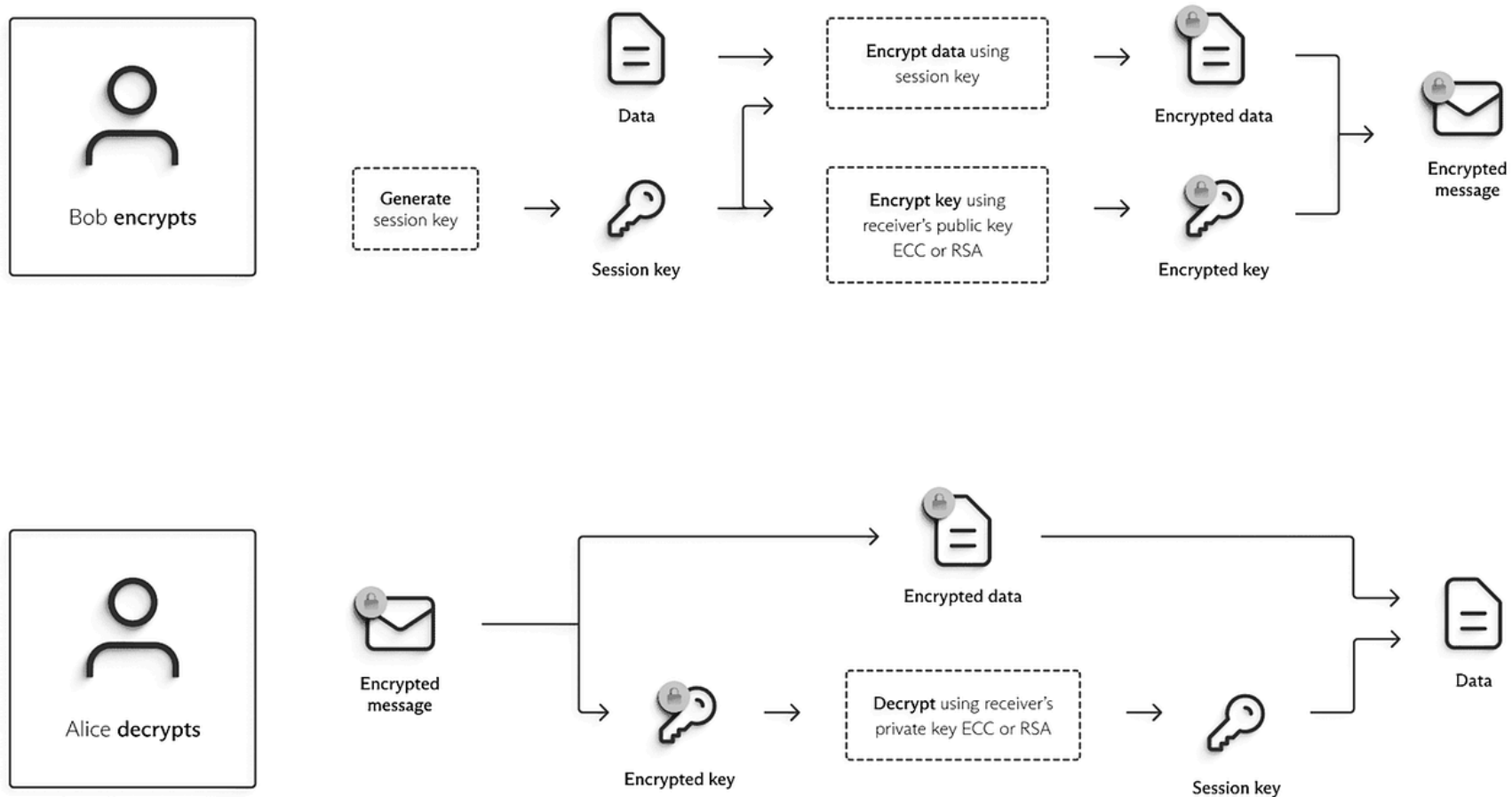


Image from: <https://medium.com/@rushikajayasinghe/what-is-pretty-good-privacy-pgp-6327e760587d>

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TL;DR: (Open)PGP is the standard, GnuPG is the *de-facto* implementation for PC.

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4. Alice sends a signed message to Carol
 - Carol is able to decrypt it and sees the **indirectly trusted** (undefined) signature

Web of Trust

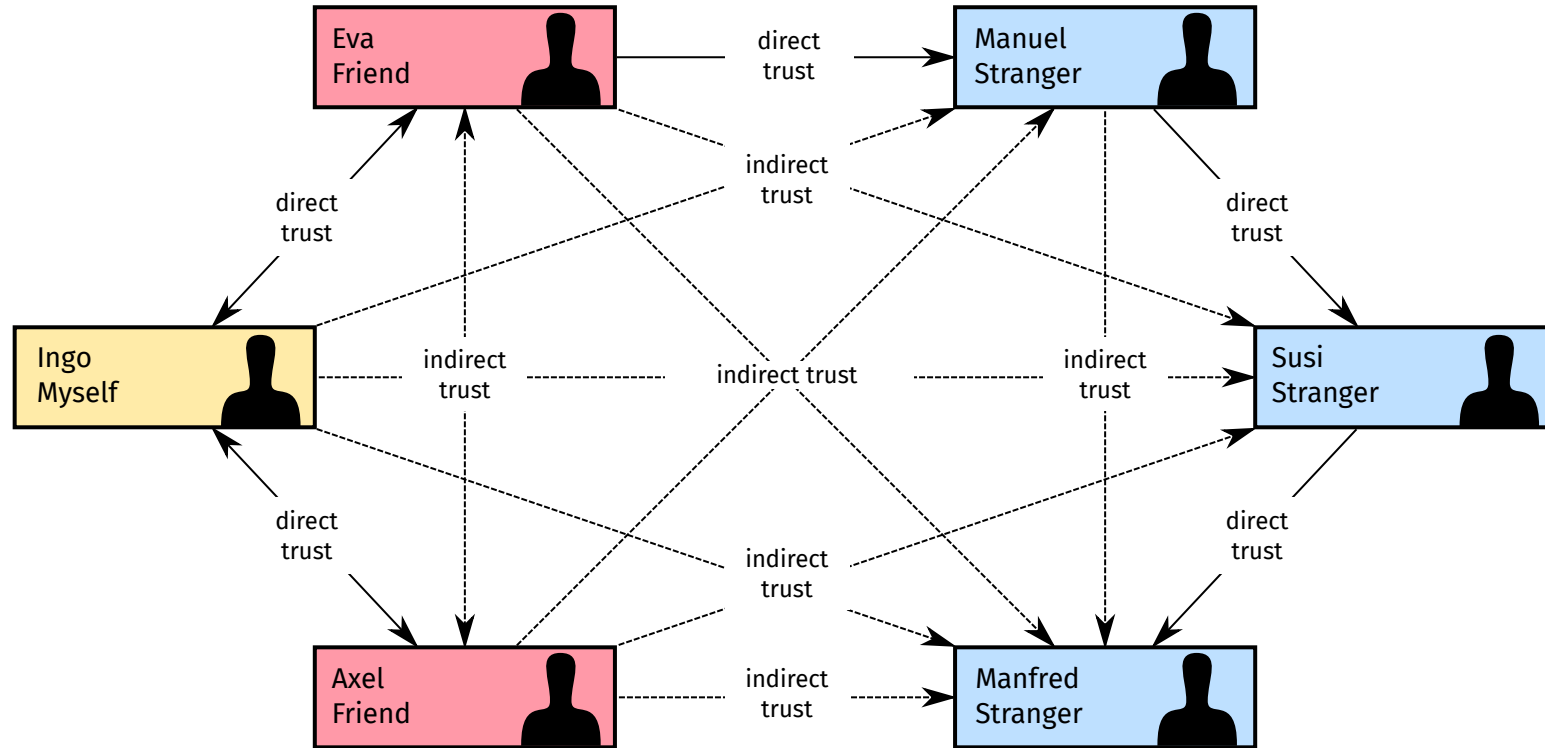
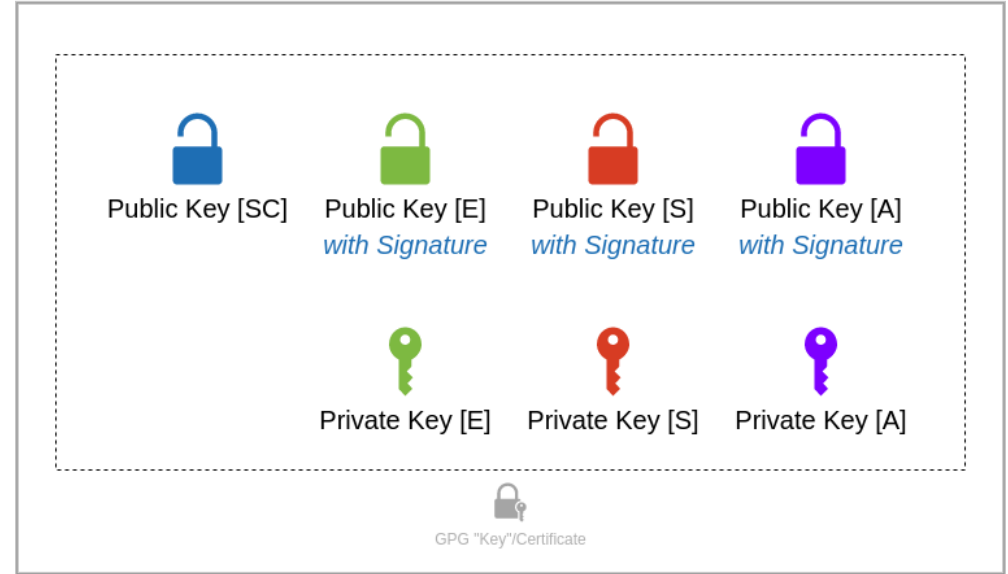


Figure 1: By Kku - Own work, CC BY-SA 4.0, <https://commons.wikimedia.org/w/index.php?curid=80652637>

Hardware keys




Storage with Master Key
(used only to modify the GPG "Key")




Workstation
(used commonly for Encrypting/Signing/Authentication)

DEMO: YubiKey provisioning¹

¹Follow along: <https://github.com/lajp/YubiKey-Guide>

Enable YubiKey and authentication through it

```
services.udev.packages = [ pkgs.yubikey-personalization ];  
hardware.gpgSmartcards.enable = true;  
services.pcscd.enable = true;  
  
security.pam.services = {  
    login.u2fAuth = true;  
    sudo.u2fAuth = true;  
};  
  
security.pam.services.swaylock.text = "auth include login";
```

- FIDO2 things (in chromium)
- `systemd-cryptenroll` for enrolling LUKS keys

```
# Run sudo systemd-cryptenroll --fido2-device=auto /dev/disk/by-uuid/
ded9ed69-89cb-4415-8532-33907ffa0b1e
# to enroll yubikey as FIDO2 device
boot.initrd.luks.devices."luks-ded9ed69-89cb-4415-8532-33907ffa0b1e" = {
    device = "/dev/disk/by-uuid/ded9ed69-89cb-4415-8532-33907ffa0b1e";
    crypttabExtraOpts = [
        "fido2-device=auto"
    ];
};
```

- age with piv: <https://github.com/str4d/age-plugin-yubikey>

- <https://github.com/drduh/YubiKey-Guide>
- <https://web.archive.org/web/20251027132314/https://rgoulter.com/blog/posts/programming/2022-06-10-a-visual-explanation-of-gpg-subkeys.html>
- <https://medium.com/@rushikajayasinghe/what-is-pretty-good-privacy-pgp-6327e760587d>
- <https://users.ece.cmu.edu/~adrian/630-f04/PGP-intro.html>